

PostgreSQL

Class 01

Introduction to Databases & PostgreSQL

- **What is Data & Database?**
- **DBMS vs RDBMS**
- **What is PostgreSQL?**
- **Types of postgresQL Commands**
- **Examples of RDBMS software**

What is DBMS?

A database is like a digital brain 🧠.

It keeps information of all the movies and series you watched on netflix.

So, that you can search,update,reterive it easily.

What is DBMS?

- A database is a structured collection of information, usually stored in tables with rows and columns.
- It helps in storing, managing, and updating large amounts of data in an organized way.
- It allows quick access and retrieval of information, making it useful for applications like banking, shopping, or social media.

What is RDBMS?

A Relational Database Management System (RDBMS) is software that stores data in structured tables with rows and columns, organizes relationships using keys, and allows efficient storage, retrieval, and management of large amounts of information.

Here are some popular **examples of RDBMS software**:

- **MySQL**
- **PostgreSQL**
- **Oracle Database**
- **Microsoft SQL Server**
- **IBM Db2**

What is PostgreSQL?

PostgreSQL is an **open-source relational database management system (RDBMS)**.

It is widely used for:

- Backend development
- Data analysis
- Enterprise applications

PostgreSQL is known for:

- High performance
- Data integrity
- Advanced SQL support

Why postgresSQL?

PostgreSQL is one of the most **powerful and trusted relational databases** used in the industry.

Open-Source & Free

- Completely **free to use**
- No license cost
- Strong community support
- Used by startups and large enterprises

No vendor lock-in.

Reliable & ACID Compliant

PostgreSQL follows **ACID** principles:

- **Atomicity**
- **Consistency**
- **Isolation**
- **Durability**

This means:

- Data is safe
- Transactions are reliable
- No data corruption

Advanced SQL Support

PostgreSQL supports:

- Complex joins
- Subqueries
- Window functions
- CTEs (WITH clause)
- Triggers & stored procedures

Best database for **learning real SQL**.

High Performance & Scalability

- Handles large datasets
- Supports indexing, partitioning
- Optimized query planner

Used for:

- Web apps
- Analytics systems
- Enterprise applications

Extensible & Flexible

You can:

- Add custom data types
- Write custom functions
- Use extensions (PostGIS, pg_trgm)

Example:

- **PostGIS** → GIS & location-based data

JSON & NoSQL Support

PostgreSQL supports:

- JSON
- JSONB

This allows:

- Relational + semi-structured data together
- Modern app development

Cross-Platform

Runs on:

- Windows
- macOS
- Linux
- Cloud platforms (AWS, Azure, GCP)

Industry Adoption

Used by:

- Instagram
- Spotify
- Netflix
- Uber
- Reddit

Trusted at scale.

Best Choice for Students & Data Roles

PostgreSQL is ideal for:

- Data Analysts
- Backend Developers
- Data Engineers
- SQL Learners

Skills are **highly transferable**.

Downloading MySQL

- Open a web browser
- Go to the **official PostgreSQL website** <https://www.postgresql.org/>
- Navigate to the **Download** section
- Select your operating system:
 - Linux
 - macOS
 - Windows

Always download PostgreSQL from the **official website** to avoid security issues.

Step 2: Choose PostgreSQL Version

- Select the **latest stable version**
- Recommended for beginners:
 - PostgreSQL 14 / 15 / 16 (depending on availability)

Why latest version?

- Better performance
- New features
- Security fixes

Step 3: Download Installer

For Windows

- Click **Download the installer**
- This uses the **EDB (EnterpriseDB) Installer**
- .exe file will be downloaded

For macOS

- Download .dmg installer
OR
- Use Homebrew (advanced users)

For Linux

- Use package manager:
 - apt (Ubuntu)
 - yum / dnf (RedHat-based)

Step 4: Install PostgreSQL (Windows Installer Flow)

1. Run the downloaded .exe file
2. Click **Next** to start installation
3. Choose installation directory
4. Select components:
 - PostgreSQL Server
 - pgAdmin 4
 - Command Line Tools

Keep default selection (recommended)

Step 5: Set Database Password

- Set a **strong password** for:
 - `postgres` (default superuser)
- This password is important:
 - Used to log in
 - Used in pgAdmin
 - Used by applications

Do not forget this password.

Step 6: Set Port Number

- Default port: **5432**
- Recommended: keep default

PostgreSQL uses this port to accept connections.

Step 7: Choose Locale

- Select default locale
- Used for sorting & character handling

For beginners:

Keep default option

Step 8: Complete Installation

- Click **Next**
- Click **Install**
- Wait for installation to finish
- Click **Finish**

PostgreSQL is now installed on your system.

Step 9: Verify Installation

Using pgAdmin

1. Open **pgAdmin 4**
2. Enter the password you set
3. Connect to PostgreSQL server

If pgAdmin opens successfully → installation is correct

Using Command Line

Open terminal / command prompt and run:

```
psql --version
```

If version appears → PostgreSQL is installed correctly